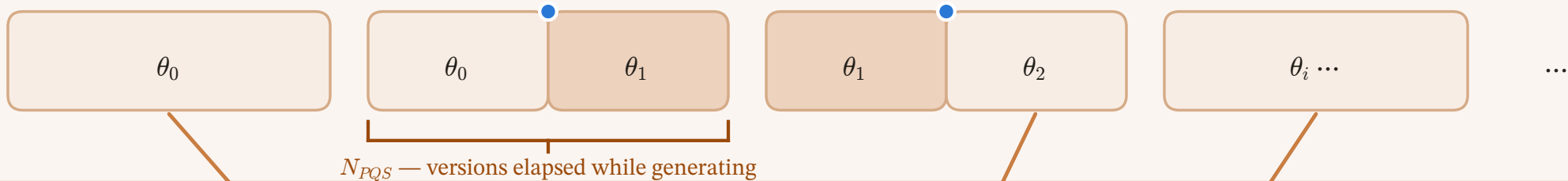


Rollout engine — SGLang

never waits for the trainer; new weights land mid-generation



FIFO rollout queue (capacity Q)



N_{IQS} — versions elapsed while waiting in the queue

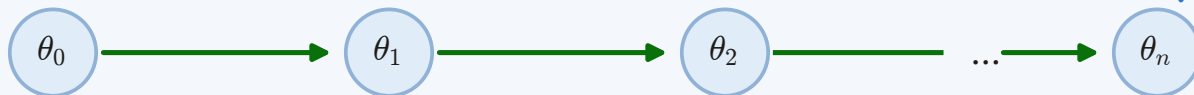
pull B = G·S rollouts whenever the queue has enough

$N = N_{PQS} + N_{IQS}$ is now a random variable across tokens

update_weights()
hot-swapped into
running engines

Training engine — Megatron

pulls from the queue; never waits for a full cadence



rollout data flow update_weights() hot swap optimizer steps weights land mid-generation